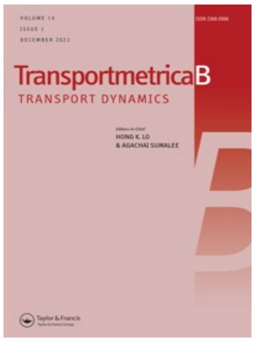

Transportmetrica B: Transport Dynamics

ISSN: 2168-0566 (Print) 2168-0582 (Online) Journal homepage: www.tandfonline.com/journals/ttrb20

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To cite this article: Néstor García-Chan, Lino J. Alvarez-Vázquez , Aurea Martínez & Miguel E. Vázquez-Méndez (2026) A mathematical approach for designing speed-limit and low/zero emission zones in urban areas, Transportmetrica B: Transport Dynamics, 14:1, 2718458, DOI: 10.1080/21680566.2026.2718458

To link to this article: https://doi.org/10.1080/21680566.2026.2718458

Published online: 18 Aug 2026.

TRANSPORTMETRICA B: TRANSPORT DYNAMICS
2026, VOL. 14, NO. 1, 2718458
https://doi.org/10.1080/21680566.2026.2718458

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RESEARCH ARTICLE

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A mathematical approach for designing speed-limit and low/zero emission zones in urban areas

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ABSTRACT

Urban policies like Low-Emission Zones (LEZs) and Zero-Emission Zones (ZEZs) are crucial for accelerating pollutant reduction. This paper introduces a numerical multi-objective optimization framework to determine the optimal spatial extent of these restricted areas by balancing environmental benefits and economic costs. The methodology integrates the finite element method with a semi-implicit time scheme to efficiently compute coupled traffic and emission dynamics. We address two scenarios: a single-control problem identifying the optimal radius of a Zone 30 via exhaustive search, and a multi-control problem defining a combined LEZ (outer annulus) and ZEZ (inner disk). The latter employs a hybrid Genetic Algorithm and Pattern Search to compute the Pareto front. Applied to the Guadalajara Metropolitan Area (Mexico), this framework successfully identifies Pareto-optimal solutions, providing a robust decision-making tool to evaluate the trade-offs between air quality and economic constraints in urban planning.

ARTICLE HISTORY
Received 28 March 2026
Accepted 24 July 2026

KEYWORDS
Multi-objective optimization; pareto front; speed-limit zones; low- and zero-emission zones; macroscopic traffic flow

1 Introduction

Efforts to mitigate urban air pollution have led to diverse traffic management strategies, beginning with speed-limited areas commonly known as Zones 30. These areas define urban perimeters where the maximum speed is restricted to 30 km/h (Yannis and Michelaraki 2025) through the use of urban design elements like narrowed passages and pedestrian-priority signaling Ewing (1999). From an environmental viewpoint, these zones indirectly curb pollution by preventing aggressive acceleration and engine overexertion, thereby fostering a smoother, more constant traffic flow (Göttlich, Herty, and Ulke 2025; Int Panis, Broekx, and Liu 2006). Zones 30 represent a highly admissible entry-level strategy, as they require minimal public investment; because public services remain unchanged, parking demand and public transport are largely unaffected. Thus, they do not impose significant changes on the habits of inhabitants and vehicle owners.

Beyond speed regulation, a more logistical approach is found in Low-Emission Zones (LEZs), which act as a selective filter by restricting access to older, and high-polluting vehicles. Since their pioneering implementation in Stockholm in 1996, LEZs have spread reach a number of 363 across Europe in 2026. The benefits and limitations of LEZs have been extensively documented in the recent systematic review by Delgado-Lindeman et al. (2025), which includes ex-ante and ex-post analyses based on data science and modelling. LEZs have led to a quantifiable decrease in NO_2 and particulate matter $PM_{2.5}$ concentrations, with typical reductions of 15%–30% for NO_2 and 5%–15% for $PM_{2.5}$. This air pollution mitigation correlates directly with reduced incidences of ischemic heart disease and premature mortality. Specifically, cardiovascular and respiratory hospital admissions have decreased by 2%–7% and 3%–11%, respectively. Additionally, this mitigation is associated with a decrease in mortality from natural causes (2%–5%) and the incidence of low birth weight (2%–8%). Furthermore, these zones facilitate a reduction in chronic stress levels caused by the high-decibel environment of congested traffic.

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Despite these gains, the effectiveness of LEZs is often undermined by a well-known phenomenon called the displacement effect, as evidenced by empirical data from the London Ultra-Low Emission Zone analyzed in Hajmohammadi and Heydecker (2022). This paper reveals that while air quality improves within the restricted perimeter, non-compliant vehicles tend to cluster at the boundaries, exacerbating congestion and elevating pollutant concentrations in peripheral areas. This shift frequently necessitates larger infrastructure adaptations, such as peripheral parking, to maintain socioeconomic admissibility. Furthermore, as noted in Sarmiento, Wägner, and Zaklan (2023), the high private costs of vehicle replacement and the perceived loss of freedom can trigger significant social discontent, often outweighing the—sometimes invisible—health benefits for the average citizen.

The most radical restriction is the Zero-Emission Zone (ZEZ), which mandates the exclusive use of electric vehicles. This shift stems from evidence—such as predictive modelling for Manchester (Broster and Terzano 2025)—showing that traditional LEZs may fail to meet WHO-recommended $PM_{2.5}$ levels. According to recent ex-ante modelling for Milan (Piccoli et al. 2025), implementing a ZEZ can reduce local NO_2 and $PM_{2.5}$ concentrations by 85% and 25%, respectively, within the zone relative to 2017 levels. Furthermore, such interventions could potentially reduce mortality and morbidity indicators by up to 50% compared to the same baseline year. Also, these zones help lower stress and sleep disruptions caused by engine noise (Delgado-Lindeman et al. 2025; Piccoli et al. 2025). Despite these projected health benefits, ZEZs drastically intensify socioeconomic friction by excluding even modern hybrid cars, imposing a severe financial burden on both owners of older combustion engines and recent adopters of transitional green technologies.

It is important to note that in most of the aforementioned studies, the spatial extent of the restricted area is treated as a fixed input parameter—often dictated by existing political or administrative borders—limiting the analysis to impact assessment rather than spatial planning. While recent literature has addressed zone delimitation via multi-objective Pareto optimization (Göttlich, Herty, and Ulke 2025; Wang et al. 2025; Xue et al. 2025), these frameworks typically rely on discrete network controls, such as traffic light timings, tolling, or blocking specific arcs within a graph network. Furthermore, their state variables are frequently derived from static statistical data, microscopic models or macroscopic models (like LWR or Dynamic Traffic Assignment (DTA)) restricted to 1D road networks, which allow to capture microscopic elements as traffic lights, lane management and turning movements but them struggle to extend across a large number of streets by the unfeasible computational cost. Consequently, existing approaches remain limited in their ability to perform an optimization to reach the best restricted area which its design is inherently coupled with continuous 2D urban domains.

Macroscopic traffic modelling has evolved to alleviate the computational burden of discrete network graphs extended the LWR and DTA models to 2D continuous domains (Jiang, Ma, and Zhou 2018), but they oversimplified urban landscape by neglecting building blocks and then an important element, the turning movements. To overcome this limitation, in García-Chan et al. (2025) the city is interpreted as a 2D porous medium, establishing a rigorous physical foundation where buildings act as the solid phase that restricts traffic flow, while the streets define the fluid phase that facilitates it. Under this framework, discrete turning movements and cars discrete density are upscaled into a continuous vehicle density and a mean velocity field (conceptually representing the average zig-zag trajectory of urban driving), however others discrete elements as traffic lights or lane management were omitted. The model was later coupled with a Navier-Stokes system and a transport equation to get a wind field and the air pollutant dispersion but considering the urban landscape using the same principle of urban porous media (see García-Chan et al. (2026)). This multi-model system provides essential state variables—such as vehicle density, velocity, and CO_2 concentrations—to evaluate variations caused by changes in the urban landscape as the presence of restriction areas, wind, fleet composition or others.

We take CO_2 as the primary environmental metric rather than NO_2 or $PM_{2.5}$ due it serves as a robust proxy for the overall intensity of internal combustion activity; furthermore, since CO_2 and toxic co-pollutants share a common tailpipe origin which reduction necessarily yields a concurrent decrease in air toxicity. Consequently, the environmental objective of this framework is formulated around mitigating the mean CO_2 concentration within the restricted area.

Meanwhile, the parking flow at the boundary of the restricted perimeters is used as a robust proxy for urban socioeconomic activities. In our macroscopic representation, vehicles cannot enter the LEZ/ZEZ core freely; therefore, this flow captures the exact rate at which commuters, visitors, and shoppers transition from drivers to pedestrians to conduct their commercial and social activities within the city centre. Mathematically, accounting for this terminal flow ensures that the urban core remains accessible (Geurs and van Wee 2004), thereby preventing the economic desertification or 'dead-centre' effect often triggered by poorly planned traffic bans (Marsden 2006; Mingardo and van Meerkerk 2012). By incorporating this metric, the model establishes a realistic balance between environmental preservation and urban vitality.

Finally, a geometric simplification by characterizing the restricted area as a disk centred at the city core with a fixed radius, or as a combination of zones forming an outer annulus and an inner disk defined by two radii. It is important to note that this simplification reflects urban reality, where restricted zones are typically defined as ellipses or avoids with flattered segments and centred in the city core (as can be seen in the cases of London, Milan or Madrid (Broster and Terzano 2025; Piccoli et al. 2025; Tassinari 2024)).

Therefore, to fills the gap left by the works mentioned above this article proposes a new optimization approach to find the optimal area in the sense of Pareto optimality where traffic flow restrictions are imposed to minimize the spatio-temporal average of air pollutant concentrations within the restricted perimeters, while maximizing parking flow outside the zones as an indirect index of socioeconomic activity and an indicator of potential congestion due to insufficient parking spaces. We formulate two multi-objective optimization problems: the first aims to determine the optimal radius for a disk-shaped Zone 30, the second treats two radii as control variables to define a hybrid configuration consisting of a ZEZ (the inner disk) surrounded by a LEZ (the outer annulus). The dual-objective nature of both problems allows for the identification of Pareto-optimal radii that prevent environmental gains from being neutralized by peripheral traffic saturation.

The paper is organized as follows: Section 2 briefly yet concisely reviews the models established in García-Chan et al. (2026), covering traffic flow, emission rates, airflow, and pollutant transport. The two multi-objective control problems are formulated in Section 3, where the control variables are defined as the radii that shape the disks and annuli. In Section 4, a new semi-implicit time scheme for the traffic flow model is introduced to achieve stable solutions with reduced computational cost. This section also details the spatial and temporal discretization of the cost functionals and provides insights into the hybrid MATLAB algorithms employed for optimization. In Section 5, several numerical results regarding the optimal solutions are presented through figures illustrating average vehicle density and air pollution distributions, as well as tables summarizing the numerical values of the cost functionals for the different scenarios. Finally, concluding remarks are provided in Section 6.

2 Mathematical framework: a porous media approach

For all models described below, the computational domain $\Omega \subset \mathbb{R}^2$ comprises the urban area Ω_u and the surrounding rural-forest region Ω_s, such that $\Omega = \Omega_u \cup \Omega_s$ and $\Omega_u \cap \Omega_s = \emptyset$. The boundary Γ consists of three disjoint segments: the obstacle walls Γ_w, the wind inlet Γ_{in}, and the wind outlet Γ_{out}. Notably, specific parts of the segment Γ_w may be included or excluded depending on the requirements of each individual model.

The urban part Ω_u is modelled as a porous medium, where the area covered by streets and buildings defines the fluid and solid phases common in porous media literature (see, e.g. Das and Mukherjee, 2018). Consequently, the porosity $0 < \epsilon < 1$ represents the proportion of a reference volume occupied by streets (fluid phase), while $(1 - \epsilon)$ accounts for the portion occupied by buildings (solid phase). Four distinct physical processes are coupled within this framework: traffic flow dynamics, traffic-induced CO_2 emissions, airflow, and atmospheric transport. Given the detailed derivation of these models in García-Chan et al. (2026) and references therein.

2.1 Modelling of traffic flow and macroscopic emissions of CO_2 in a porous city

Traffic flow is governed by a modified Navier-Stokes-like system for porous media, where $\rho(\mathbf{x}, t)$ and $\mathbf{u}(\mathbf{x}, t)$ denote car density and velocity, respectively. A fundamental variable in this formulation is the instantaneous travel cost potential φ, obtained by solving an Eikonal equation (see García-Chan et al. (2025)) for further details). This variable drives the traffic dynamics through the following two key components: the desired velocity $\mathbf{u}_d$ and the traffic demand q, both of which account for the density-dependent travel cost

$f(\rho)$ within the urban domain Ω_u. First, the travel cost $f(\rho)$ is defined by accounting for congestion and speed limit $U_{\max}$ (Treiber and Kesting 2013):

$$f(\rho) = U_{\max}\left(1 - \frac{\rho}{\rho_{\max}}\right), \tag{1}$$

where $\rho_{\max}$ denotes the maximum road capacity. Consequently, the desired velocity $\mathbf{u}_d$ is defined as the product of $f(\rho)$ and an unit vector pointing in the direction of steepest descent of the potential φ, thereby directing traffic toward the destination. So, desired velocity is determined by:

$$\mathbf{u}_d = -f(\rho)\frac{\nabla\varphi}{\|\nabla\varphi\|}, \tag{2}$$

where vector ∇ stands for the classical space gradient operator.

Finally, the traffic demand q represents the rate at which vehicles are released into the flow from their parking locations. This deployment is based on a maximum capacity and is modulated by both a saturation term and the temporal dynamics of urban activity:

$$q = q_{\max}\left(1 - \frac{\varphi}{\varphi_{\max}}\right) g(t), \tag{3}$$

where $q_{\max}$ is the maximum rate, $\varphi_{\max}$ is the maximum instantaneous travel cost, and $g(t)$ accounts for daily peak and off-peak traffic hours.

In this framework, the target velocity is coupled to the momentum balance as a relaxation term, representing the drivers' tendency to adjust their current velocity $\mathbf{u}$ toward the desired direction $\mathbf{u}_d$ within a time scale τ. Simultaneously, the traffic demand q acts as the source term in a mass balance. Furthermore, the transport and dissipative properties of the traffic flow are governed by the car mass diffusivity and the macroscopic traffic viscosity. Thus, we seek for the car density ρ [veh/km^2] and the local traffic speed $\mathbf{u} = (u_1, u_2)$ [km/h] such that satisfy the governing system for the urban porous media traffic flow, being the dot $(\cdot)$ operator the time derivative:

$$\epsilon\dot{\rho} + \nabla\cdot(\rho\mathbf{u}) - \nabla\cdot(\epsilon\nu\nabla\rho) + \epsilon\kappa\rho = (1-\epsilon)q, \tag{4a}$$

$$\dot{\mathbf{u}} + \frac{1}{\epsilon}(\mathbf{u}\cdot\nabla)\mathbf{u} - c^2\rho(\nabla\cdot\mathbf{u})\mathbf{1} = \frac{\mathbf{u}_d - \mathbf{u}}{\tau} + \frac{\epsilon}{\rho}\left(\nabla\cdot\frac{\mu}{\epsilon}\nabla\right)\mathbf{u} - \left(\frac{\epsilon\mu}{\rho K} + \frac{\epsilon F}{\sqrt{K}}\|\mathbf{u}\|\right)\mathbf{u}, \tag{4b}$$

where ϵ is the urban porosity, $(1-\epsilon)q$ [veh/km^2/h] is the traffic demand function associated to the building-blocks (solid-phase), c^2 is the traffic sonic speed (computed as $c^2 = \theta/\rho_{max}$ with θ an anticipation coefficient), K is the urban permeability, F is the Forchheimer coefficient, ν [km^2/h] is the coefficient of mass diffusion, κ [h^{-1}] is the absorption (parking) rate from the streets (fluid-phase) to parking-spaces inside building-blocks (solid-phase), τ [h] is the relaxation time, and μ [km^2/h] is the viscosity.

This macroscopic model (4) considers a straight direction towards drivers' destination, but its velocity magnitude is a mean value corresponding to a zig-zag movement which depends directly on porosity. Unfortunately, microscopic elements as traffic lights, lane management or turning movements are omitted here due its microscopic nature. Also, we highlight that parameters ϵ, K, and F can be prescribed from an idealized city but also from real-data which make the model very flexible for real study cases and future validation but also for theoretical exercises as the present work.

Regarding boundary conditions, these are imposed to prevent vehicles from leaving the domain, except when they reach their destination and are removed by means of the parking flow (absorption) term on the mass balance equation. Therefore, a slip condition is imposed on the walls, and a homogeneous Neumann condition on the inlet and outlet segments.

From the macroscopic variables $(\rho, \mathbf{u})$ solution of system (4), we derive the microscopic velocity U and acceleration a by the formulas (see Jiang, Ma, and Zhou (2018)):

$$U = \|\mathbf{u}\|/\gamma_1, \tag{5a}$$

$$a = (\mathbf{a}\cdot\mathbf{u}/U)\gamma_2, \tag{5b}$$

being $\gamma_1 = 3.6$, $\gamma_2 = 10^{-3}/(3.6)^2$ factors to scale the macroscopic units [km/h^2] into microscopic ones [m/s^2], and the macroscopic acceleration $\mathbf{a}$ computed using the definition of the acceleration as the material derivative:

$$\mathbf{a} = \frac{\mathbf{u}_d - \mathbf{u}}{\tau} + c^2\rho(\nabla\cdot\mathbf{u})\mathbf{1} + \frac{\epsilon}{\rho}\nabla\cdot\left(\frac{\mu}{\epsilon}\nabla\mathbf{u}\right) - \left(\frac{\epsilon\mu}{\rho K} + \frac{\epsilon F}{\sqrt{K}}\|\mathbf{u}\|\right)\mathbf{u}, \tag{6}$$

maintaining physical consistency due is derived directly from the momentum balance (4b) as the sum of the relaxation, viscous, and porous resistance forces (see details in (Das and Mukherjee 2018; García-Chan et al. 2026)).

The quantities U and a are essential in the micro-scale emission model proposed in Int Panis, Broekx, and Liu (2006). Although this model is capable of estimating various pollutants (e.g. NO_x, PM_x or Volatile Organic Compounds (VOCs)), CO_2 is selected as the proxy of air pollution in this study. Thus, to compute the instantaneous CO_2 microscopic emissions with the regression model:

$$E_{CO_2} = \max\{f_1 + f_2U + f_3U^2 + f_4a + f_5a^2 + f_6Ua, 0.0\}, \tag{7}$$

where coefficients f_i, $i = 1, 2, \ldots, 6$ have suitable values for CO_2.

Therefore, using the scale factor γ_1 and the macroscopic density ρ we project the emissions E_{CO_2} onto emissions but in macroscopic coordinates EC by the formula proposed in García-Chan et al. (2026):

$$\mathrm{EC} = \gamma_1\rho E_{CO_2}. \tag{8}$$

2.2 Air circulation and CO_2 transport

Air circulation within our porous city obeys fluid dynamics principles similar to those of traffic flow. Specifically, the air flows through the streets (fluid-phase), while the buildings (solid-phase) acts as a stationary obstacle. Similarly, CO_2 transport is governed by advection, diffusion, and gravitational settling. These transport mechanisms occur exclusively within the fluid phase of the porous medium. Consequently, while parameters such as porosity, permeability, and Forchheimer coefficient are shared with the traffic flow model, physical differences are accounted for through specific parameters such as fluid density and diffusivity coefficient.

Within this setting, the air velocity $\mathbf{v}(\mathbf{x}, t)$ is modelled as an incompressible flow through the same porous medium, characterized by the air density ρ_a and dynamic viscosity μ_a, subject to the continuity equation as a physical constraint. Therefore, we seek a velocity vector field $\mathbf{v} = (v_1, v_2)$ [km/h] and the associated pressure field P [kg/km/h^2] that satisfy the following system:

$$\dot{\mathbf{v}} + \frac{1}{\epsilon}(\mathbf{v}\cdot\nabla)\mathbf{v} + \frac{\epsilon}{\rho_a}\nabla P = \frac{\epsilon}{\rho_a}\nabla\cdot\left(\frac{\mu_a}{\epsilon}\nabla v\right) - \left(\frac{\epsilon\mu_a}{\rho_a K} + \frac{\epsilon F}{\sqrt{K}}\|\mathbf{v}\|\right)\mathbf{v}, \tag{9a}$$

$$\nabla\cdot\mathbf{v} = 0. \tag{9b}$$

Furthermore, given the macroscopic emission concentration EC from Equation (8) as the CO_2 source term, and adequate coefficients for diffusivity μ_ϕ and deposition rate σ, the advection-diffusion-reaction model governing the carbon dioxide pollutant concentration $\phi(\mathbf{x}, t)$ [kg/km^2] is formulated as:

$$\epsilon\dot{\phi} + \mathbf{v}\cdot\nabla\phi - \nabla\cdot(\epsilon\mu_\phi\nabla\phi) + \epsilon\sigma\phi = \zeta EC, \tag{10}$$

where ζ is a heterogeneous coefficient that accounts for the emissions from combustion, hybrid, and electric vehicles, as detailed below (see Equation (16)).

Boundary conditions for both models are prescribed to ensure that air enters through the inlet segment Γ_{in} and exits via the outlet segment Γ_{out}, while being diverted around solid obstacles. These conditions facilitate the evacuation of pollutants through the outlet part of the boundary, prevent any leakage through the walls Γ_w, and ensure that the only pollution source is the traffic flow within the domain interior (a detailed description is provided in García-Chan et al. (2026)).

3 Optimal design of speed-limit, low- and zero-emissions zones

In this section, we formulate two optimal control problems based on environmental traffic management strategies currently deployed by local governments. The first intervention focuses on speed-limit zones (commonly known as Zones 30), where maximum speeds are strictly capped to reduce tailpipe emissions, requiring only traffic-calming measures and leaving unchanged the parking capacities. The second strategy combines Low Emission Zones (LEZs) and Zero Emission Zones (ZEZs). This policy enforces a graduated prohibition scheme where internal combustion vehicles are banned from both zones, while hybrid vehicles are additionally excluded from the innermost ZEZ, effectively reserving the city core for purely electric mobility. In contrast with speed-limit zones this strategy requires large investments and could generate social obstruction as was mentioned above.

Both interventions are modelled within a multi-objective framework aimed at simultaneously minimizing average pollutant concentrations within the restricted perimeters (J_P) and maximizing the vehicle flow toward parking infrastructure (J_K) as a proxy of a socioeconomic rate at which commuters, visitors, and shoppers conduct their commercial and social activities within the city centre. Following the ellipsoid or ovoid shape of real LEZ and ZEZ our restricted zones are modelled using a concentric circular layout. This geometric formulation closely reflects the mono-centric urban sprawl and radial traffic patterns typical of emerging metropolitan areas like Guadalajara (Guerra 2015; Monkkonen and Comandon, 2018), while aligning with the roughly elliptical or ovoid structures centred on the city core observed in European implementations like Paris (Poulhès, Proulhac, and Malriat 2025) and London (Broster and Terzano 2025).

Formally, we consider the multi-objective optimization problem: Find an optimal radii vector $\mathbf{r} \in \mathbb{R}^n$ such that satisfies the following problem:

$$\min_{\mathbf{r}\in R} \mathbf{J}(\mathbf{r}) = (J_1(\mathbf{r}), J_2(\mathbf{r}), \ldots, J_k(\mathbf{r})), \tag{11}$$

where R is the feasible set of control radii, $\mathbf{r}$ is a vector of controls, and each $J_i(\mathbf{r})$ is an objective to minimize. Since a single solution that simultaneously minimizes all objectives rarely exists, we seek solutions in the sense of Pareto optimality:

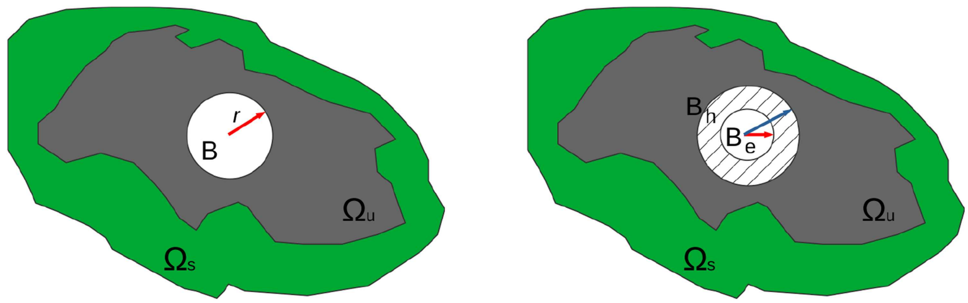
Definition 3.1. (Pareto optimality) A solution $\mathbf{r}^* \in R$ is said to be Pareto-optimal if there is no other $\mathbf{r} \in \mathcal{R}$ such that $J_i(\mathbf{r}) \le J_i(\mathbf{r}^*)$ for all $i \in \{1, \ldots, k\}$ and $J_j(\mathbf{r}) < J_j(\mathbf{r}^*)$ for at least one index j. The set of all Pareto-optimal solutions is known as the Pareto set, and its projection onto the objective space is the Pareto front.

While for $k \le 3$ the Pareto front can be fully characterized via any appropriate search method, a higher dimension $k \ge 4$ makes the Pareto front impractical to visualize. Thus, a priori or interactive techniques (Hillermeier 2012; Miettinen 1999) are commonly used to find a specific front point (or compromise solution).

3.1 Optimal control problem 1: speed-limit zone

Numerical experiments in García-Chan et al. (2026) suggest that implementing maximum speed limits can be an effective strategy to protect sensitive areas from high concentrations of air pollutants, such as CO_2. In this study, we formulate an optimal control problem to delimit a zone $\mathrm{B} \subset \Omega_u$, defined as a disk with radius $r > 0$ centred in the city core, where the maximum speed is restricted to 30 km/h (see Figure 1a). To evaluate this strategy, we define the two following objective functionals: first, the average CO_2 concentration, given by the integral:

$$J_P(r) = \frac{1}{|\mathrm{B}|T} \int_0^T \int_{\mathrm{B}} \phi(\mathbf{x}, t)\, \mathrm{d}\mathbf{x}\mathrm{d}t, \tag{12}$$



(a) The speed-limit zone as a disk B of radius r.

(b) The LEZ as the outer annulus and the ZEZ as the inner disk.

Figure 1. Design of restricted zones in the urban domain Ω_u and its rural surroundings Ω_s: (a) The speed-limit zone defined as a disk B with centre in the city core and radius r. (b) The LEZ as an outer annulus $\mathrm{B_h}\backslash\mathrm{B_e}$ of thickness $(r_h - r_e)$ banned to internal combustion vehicles, and the ZEZ as an inner disk $\mathrm{B_e}$ with radius r_e banned to hybrid vehicles.

where $|\mathrm{B}|$ denotes the area of B.

Also, assuming a pre-existing distribution of parking spaces with a specific parking rate $\kappa(\mathbf{x})$ and infinite capacity, and aiming to maximize the parking flow, we define the related objective functional as:

$$J_K(r) = \frac{1}{|\mathrm{B}|T}\int_0^T \int_{\mathrm{B}} \kappa(\mathbf{x})\rho(\mathbf{x}, t)\mathrm{d}\mathbf{x}\mathrm{d}t. \tag{13}$$

Consequently, our goal is to mitigate air pollution levels while simultaneously maximizing the traffic flow into parking facilities within the sensitive zone $\mathrm{B} \subset \Omega_u$. To achieve this, we seek an optimal radius r by solving the following optimal control problem:

$$\begin{cases} \text{Minimize} & \mathbf{J}(r) = (J_P(r), -J_K(r)) \\ \text{subject to} & r_{lw} \le r \le r_{up} \end{cases}, \tag{14}$$

where r_{lw} and r_{up} represent the lower and upper bounds, respectively. These constraints are determined by construction costs, land use, changes in citizen habits, or other logistical barriers to implementing speed limits. Given the single-control nature of the problem (varying only r), its solution is obtained in a straightforward manner through an exhaustive computation of both cost functionals, allowing us for determining the entire Pareto front, which provides a complete map of the trade-offs between air quality and commercial accessibility. This stands for this simple case in contrast to the following, more complex scenario.

3.2 Optimal control problem 2: LEZ and ZEZ design

We seek to determine two optimal radii r_e and r_h, $r_e < r_h$, defining the disks $\mathrm{B_e}$ and $\mathrm{B_h}$, which delimit the Unrestricted Zone (URZ) as the subdomain $\Omega_u\backslash\mathrm{B_h}$ where are allowed any kind of vehicle, the LEZ as the outer annulus $\mathrm{B_h}\backslash\mathrm{B_e}$ of thickness $(r_h - r_e)$ as a banned zone for internal combustion cars, and the ZEZ defined as the inner disk $\mathrm{B_e}$ itself as a banned zone for combustion and hybrid cars (see Figure 1b). Also, car density ρ must be decomposed into combustion, hybrid, and electric cars categories using the proportions p_c, p_h and p_e, such that $p_c, p_h, p_e \ge 0$ and $p_c + p_h + p_e = 1$. To do it, we define a travel cost $f(\rho)$ formulated as:

$$f(\rho) = U_{max}\left[\left(1 - \frac{\rho}{\rho_{max}}\right)\mathbf{1}_{\mathrm{URZ}} + \left(1 - \frac{\rho}{(p_e + p_h)\rho_{max}}\right)\mathbf{1}_{\mathrm{LEZ}} + \left(1 - \frac{\rho}{p_e\rho_{max}}\right)\mathbf{1}_{\mathrm{ZEZ}}\right], \tag{15}$$

being $\mathbf{1}_\omega$ the indicator function of any subdomain $\omega \subset \Omega$.

Similarly, CO_2 emissions are modelled via an heterogeneous coefficient ζ, accounting for the vehicle stock in each banned zone:

$$\zeta = \begin{cases} p_c + \frac{p_h}{2}, & \text{in URZ}, \\ \frac{p_h}{2}, & \text{in LEZ}, \\ 0, & \text{in ZEZ}. \end{cases} \tag{16}$$

This formulation assumes that emissions from hybrid vehicles are half of those produced by internal combustion engines, while electric vehicles are considered zero-emission. Following García-Chan et al. (2026), the resulting emission field must be directly proportional to this new coefficient ζ. Thus, the source term is now defined as ζEC, where EC is obtained from Equation (8).

The first cost objective J_P, to be minimized, is now the spatial-temporal average of the CO_2 concentration within the hybrid banned zone $\mathrm{B_e}$. The second one, J_K, corresponds here to the average parking flow within the combustion zone $\Omega_u \backslash \mathrm{B_h}$, which we aim to maximize; this is, for $\mathbf{r} = (r_h, r_e)$:

$$J_P(\mathbf{r}) = \frac{1}{|\mathrm{LEZ}|\,T} \int_0^T \int_{\mathrm{LEZ}} \phi(\mathbf{x}, t) \mathrm{d}\mathbf{x} \mathrm{d}t, \tag{17}$$

$$J_K(\mathbf{r}) = \frac{1}{|\mathrm{URZ}|\,T} \int_0^T \int_{\mathrm{URZ}} \kappa(\mathbf{x}) \rho(\mathbf{x}, t) \mathrm{d}\mathbf{x} \mathrm{d}t. \tag{18}$$

Additionally, we introduce a new vector cost functional $\mathbf{G}$ that can model the implementation costs of the restricted zones, such as parking infrastructure, public transport, electric vehicle (EV) charging stations, signage, and street adaptations. The socioeconomic costs associated with these urban transformations vary by zone. Indeed, the combustion vehicle zone requires a higher density of parking spaces and public transport facilities near the boundary defined by r_h to expedite drivers access to the city centre.

By defining a fixed, adequate increment $\Delta_p = r_p - r_h$, the outermost boundary radius r_p becomes a dependent variable $r_p = r_h + \Delta_p$. Thus, we can compute the area of the annulus where these significant modifications must be implemented as $A_p(r_h) = \pi[(r_h + \Delta_p)^2 - r_h^2]$, with its corresponding socioeconomic cost given by $G_p = E_p A_p$.

Similarly, we compute the area for hybrid and electric cars as $A_h(r_h, r_e) = \pi(r_h^2 - r_e^2)$, and the exclusive EV area as $A_e(r_e) = \pi r_e^2$, along with their respective implementation costs $G_h = E_h A_h$ and $G_e = E_e A_e$. In both inner areas, the interventions require fewer parking spaces and electric infrastructure, proportional to a smaller vehicular fleet. Therefore, the constant values E_p, E_h, E_e are estimated to represent the macroscopic cost per square kilometre in each zone. Thus, the vector cost functional $\mathbf{G}$, evaluated strictly over the decision variables (r_h, r_e), is formulated as:

$$\mathbf{G}(\mathbf{r}) = (G_p(r_h), G_h(r_h, r_e), G_e(r_e)), \tag{19}$$

assuming that $E_p \gg E_h > E_e$, due to the greater need for parking and transit infrastructure required for the more peripheral area.

To establish the layout of the zero-emission zone and the restricted zone for internal combustion cars while maintaining socioeconomic activity (measured by parking flow), we solve the following constrained multi-objective optimal control problem. We seek the optimal pair of radii $\mathbf{r} = (r_h, r_e)$ such that:

$$\begin{cases} \text{Minimize} & \mathbf{J}(\mathbf{r}) = (J_P(\mathbf{r}), -J_K(\mathbf{r}), G_p(r_h), G_h(r_h, r_e), G_e(r_e)) \\ \text{subject to} & r_h - r_e \geq \Delta r, \\ & r_{h,lw} \leq r_h \leq r_{h,up}, \\ & r_{e,lw} \leq r_e \leq r_{e,up}, \end{cases} \tag{20}$$

where Δr, $r_{h,lw}$, $r_{h,up}$, $r_{e,lw}$ and $r_{e,up}$ represent suitable constraint bounds.

In this case, since the number of objectives is high (greater than three), generating the complete Pareto front is not appropriate, not only because it is computationally expensive, but also because it is hard for the decision maker (the local government) to select from a large set of alternatives (Miettinen 1999). Then, as a first approach to the problem, we think of an a priori strategy, assuming that there are available weight vectors $\boldsymbol{\lambda}_W = (\lambda_P, \lambda_K, \lambda_p, \lambda_h, \lambda_e)$ with $\lambda_j \geq 0$ and $\sum \lambda_j = 1$, which represent the preferences of the decision-maker and allow us to seek for specific points on the Pareto front (specific Pareto solutions). According to this assumption, the multi-objective problem (20) is transformed into a single-objective one by minimizing the weighted sum:

$$\begin{cases} \text{Minimize} & \boldsymbol{\lambda}_W \cdot \mathbf{J}(\mathbf{r}) \\ \text{subject to} & r_h - r_e \geq \Delta r, \\ & r_{h,lw} \leq r_h \leq r_{h,up}, \\ & r_{e,lw} \leq r_e \leq r_{e,up}, \end{cases} \tag{21}$$

being the dot operator $(\cdot)$ the usual inner product between vectors. By varying the combinations in $\boldsymbol{\lambda}_W$, we can identify 'extreme cases' (prioritizing only one objective) and 'compromise solutions' (balancing environmental health with urban implementation costs).

4 Numerical solution of state variables and the optimal control problems

The multi-objective problems (14)–(20) are nonconvex; therefore, multiple local optima are expected. Furthermore, because the control variables are not explicitly formulated within the problem, gradient computation is infeasible. This leads us to employ derivative-free methods. However, these methods require not only an efficient numerical scheme to evaluate the objective functions but also stable and efficient numerical solutions of models (4), (9) and (10) in order to obtain its state variables. All models were solved numerically in previous authors' papers (see (García-Chan et al. 2025, 2026; García-Chan, Licea-Salazar, and Gutierrez-Ibarra 2024)), using explicit schemes for time integration in the traffic flow model, and semi-implicit ones in case of the others. The explicit scheme is not adequate in a context of solving optimal control problems which requires numerous evaluations of the cost functionals. Consequently, this section begins by formulating a new semi-implicit scheme for the traffic flow model, followed by a brief description of the finite element method used for air velocity and CO_2 transport. Finally, we detail the discretization of the cost functionals and the implementation of a hybrid solver that combines a genetic algorithm with a derivative-free method to efficiently solve the discrete optimal control problems.

4.1 A semi-implicit scheme to solve the traffic flow model

This section introduces an efficient numerical framework for solving the traffic flow model (4). To accommodate the high number of cost functional evaluations required by optimization solvers we formulate a semi-implicit α-family scheme with the aim to increase the time step Δt. Thus, we define $\Delta t = T/N$ and the discrete time steps as $t^n = n\Delta t$; for a given state variable ψ its intermediate state is formulated as $\psi^{n+\alpha} = (1-\alpha)\psi^n + \alpha\psi^{n+1}$ where $\alpha \in [0.5, 1]$ to ensure at least first-order accuracy and improved stability. Therefore, the resulting semi-discrete problem for system (4) is formulated as: Given an initial state for density $\rho^0 = \rho(\cdot,0)$ and velocity $\mathbf{u}^0 = \mathbf{u}(\cdot,0)$, we seek for ρ^{n+1} and $\mathbf{u}^{n+1}$ such that satisfies for $n = 0, \ldots, N,$, the system:

$$\epsilon\frac{\rho^{n+1}-\rho^n}{\Delta t} + \nabla\cdot(\rho^{n+\alpha}\mathbf{u}^n) - \nabla\cdot(\epsilon\nu\nabla\rho^{n+\alpha}) + \epsilon\kappa\rho^{n+\alpha} = (1-\epsilon)q^{n+\alpha}, \tag{22a}$$

$$\begin{aligned} &\frac{\mathbf{u}^{n+1}-\mathbf{u}^n}{\Delta t} + \frac{1}{\epsilon}(\mathbf{u}^n\cdot\nabla)\mathbf{u}^{n+\alpha} - c^2\rho^{n+1}(\nabla\cdot\mathbf{u}^{n+\alpha})\mathbf{1} + \frac{\mathbf{u}_d^{n+1}-\mathbf{u}^{n+\alpha}}{\tau} = \\ &\frac{\epsilon}{\rho^{n+1}}\nabla\cdot\left(\frac{\mu}{\epsilon}\nabla\mathbf{u}^{n+\alpha}\right) - \left(\frac{\epsilon\mu}{\rho^{n+1}K} + \frac{\epsilon F}{\sqrt{K}}\|\mathbf{u}^n\|\right)\mathbf{u}^{n+\alpha}. \end{aligned} \tag{22b22b}$$

This semi-discrete formulation decouples velocity $\mathbf{u}^n$ from density (splitting approach), allowing the continuity equation to be solved first for ρ^{n+1}, followed by the instantaneous travel cost $\varphi(\rho^{n+1})$ obtaining the update of the desired velocity $\mathbf{u}_d^{n+1}$, and then the momentum equation for $\mathbf{u}^{n+1}$. This decoupling is physically justified for the subsonic regime of the traffic flow, where a small pressure wave speed weakens the pressure term. While the use of previous step velocity $\mathbf{u}^n$ technically reduces the formal temporal accuracy, this consistency error remains negligible compared to the significant gains in computational efficiency (Logg, Mardal, and Wells 2012). Furthermore, the high viscosity of the model acts as a natural damping mechanism that enhances the robustness of the linear solvers allowing stable simulations at much larger time steps. The spatial discretization is performed using a Galerkin Finite Element Method with the space V_h and the vector space $\mathbf{W}_h$ composed of continuous piecewise linear functions (P_1). For the sake of brevity, the variational form of the system (22), as well as the specific details on integration by parts and boundary condition implementation, are omitted here as they follow the procedures detailed in García-Chan et al. (2025, 2026). The resulting nonlinear algebraic system is addressed using a fixed-point (Picard) iteration strategy. By linearizing the advective and Forchheimer terms, and the stabilizing effect given by the viscosity terms our approach achieves convergence within 3 to 5 iterations per time step in the numerical experiments, maintaining the efficiency of the semi-implicit scheme.

4.2 Numerical framework for airflow and transport models

The governing equations for air velocity (9) and pollutant transport (10) are solved using a numerical framework based on the finite element method, semi-implicit time schemes and the Galerkin-Least-Squares stabilization. For further details on these schemes, the reader is referred to our previous works (García-Chan et al. 2025, 2026; García-Chan, Licea-Salazar, and Gutierrez-Ibarra 2024).

4.3 Computational assessment of cost functionals

To evaluate the cost functionals over the spatial mesh is important to emphasize here that our framework utilizes two distinct meshes: one for traffic flow and another for air pollution. The latter is more refined to cover areas where vehicle access is prohibited, such as parks. This dual-mesh approach requires careful indexing in the discrete integrals. Let nt_P and nd_P denote the number of triangles and nodes in the air pollution mesh, respectively. Similarly, let nt_K and nd_K represent the corresponding values for the traffic flow mesh. For problem (14), given a control radius r and the corresponding restricted velocity zone $\mathrm{B} \subset \Omega_u$, we denote by i the indices of triangles in the pollution mesh belonging to B, and by j those in the traffic mesh. If $|\mathcal{T}^e|$ represents the area of a triangular element $\mathcal{T}^e$, the discrete area $|\mathrm{B_h}|$ is approximated by $\sum_i |\mathcal{T}^i|$ or $\sum_j |\mathcal{T}^j|$, depending on the mesh being used.

Now, given the discrete functions for air pollution concentration $\{\{\phi_l^n\}_{l=1}^{nd_P}\}_{n=0}^{N}$, car density $\{\{\rho_m^n\}_{m=1}^{nd_K}\}_{n=0}^{N}$, and parking rate $\{\kappa_m\}_{m=1}^{nd_K}$ we evaluate the discrete cost functionals as:

$$J_P^{\Delta} = \frac{1}{T|\mathrm{B_h}|} \sum_{T^i \subset B} \frac{|T^i|}{3} \left[\sum_{\mathbf{x}_l \in T^i} \frac{\Delta t}{2} \left(\phi_l^0 + \phi_l^N + 2 \sum_{n=1}^{N-1} \phi_l^n \right) \right], \tag{23a}$$

$$J_K^{\Delta} = \frac{1}{T|\mathrm{B_h}|} \sum_{T^j \subset B} \frac{|T^j|}{3} \left[\sum_{\mathbf{x}_m \in T^j} \frac{\Delta t}{2} \kappa_m \left(\rho_m^0 + \rho_m^N + 2 \sum_{n=1}^{N-1} \rho_m^n \right) \right]. \tag{23b}$$

This choice of numerical integration is consistent with the P_1 finite element approximation used in our solvers. Since the state variables ϕ and ρ are piecewise linear over the triangular elements, the spatial quadrature is exact. Furthermore, the trapezoidal rule provides a second-order accurate integration, ensuring that the discrete cost functionals J_P^{Δ} and J_K^{Δ} are smooth. This smoothness is critical for the stability of the optimization solvers, which demand a great number of cost functional evaluations.

The integrals for the cost functional in problem (21) are computed analogously, considering the outer annulus $\mathrm{B_h} \backslash \mathrm{B_e}$ = LEZ and the inner disk $\mathrm{B_e}$ = ZEZ, but their detailed formulas are omitted here for brevity.

To solve the problem (17), its single control r allows for determining the entire Pareto front through an exhaustive computation of both cost functionals. In contrast, due to the high number of objectives, problem (20) is addressed with an a priori strategy, and only some Pareto solutions are obtained by solving the single-optimization problem (21) within the two-dimensional control space (r_h, r_e). Given the likely nonconvexity of the cost functional and the lack of gradient information, we implement a hybrid optimization approach. First, a Genetic Algorithm (GA) explores the admissible set to identify promising candidate regions mitigating the risk of entrapment in local minima. Subsequently, the derivative-free solver Pattern Search (PS) is used to refine this initial guess and converge toward a high quality solution.

Since GA and PS methods require hundreds of cost functionals evaluations, the large-scale sparse matrices from our system were offloaded to the GPU (Cecka, Lew, and Darve 2011; Li and Saad 2013). The linear systems contained in the fixed-point loop are solved via the Preconditioned Conjugate Gradient (PCG) method using an ILUTP preconditioner (Saad 2003) resulting in a computationally feasible methodology.

5 Numerical experiments

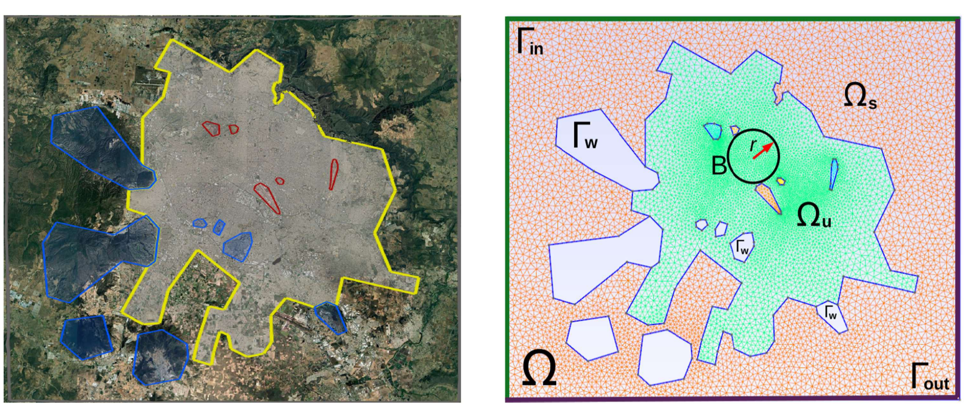
In this section, we present the initial distribution of state variables and given data required by the numerical solution of the optimal control problems formulated in Sections (3.1) and (3.2), and the numerical results achieved for them.

5.1 Case study: computational domain based on the Guadalajara Metropolitan Area (Mexico)

The rectangular computational domain Ω encompasses both the Guadalajara Metropolitan Area (GMA) and its surrounding rural zones (see Figure 2a). To ensure consistency across the different physical models, we utilize a nested polygonal discretization (García-Chan et al. 2026) generated with the license free software Gmsh (Geuzaine and Remacle 2009). This approach allows the creation of a modular mesh where specific urban features –such as industrial zones, parks, and university campuses– can be either included or omitted depending on the simulation requirements. Specifically, the airflow and pollutant transport models employ a refined mesh (9363 nodes, 18,258 triangles) to account for these internal features (see Figure 2b), while the traffic flow model uses a simplified version by omitting zones restricted to vehicles (not shown here for the sake of conciseness). Finally, the domain boundary is partitioned into three functional segments: inlet Γ_{in}, outlet Γ_{out}, and natural obstacles Γ_w, as illustrated in Figure 2b. These segments facilitate the imposition of specific boundary conditions for each model –traffic flow, wind velocity, and pollutant transport– ensuring physical consistency.

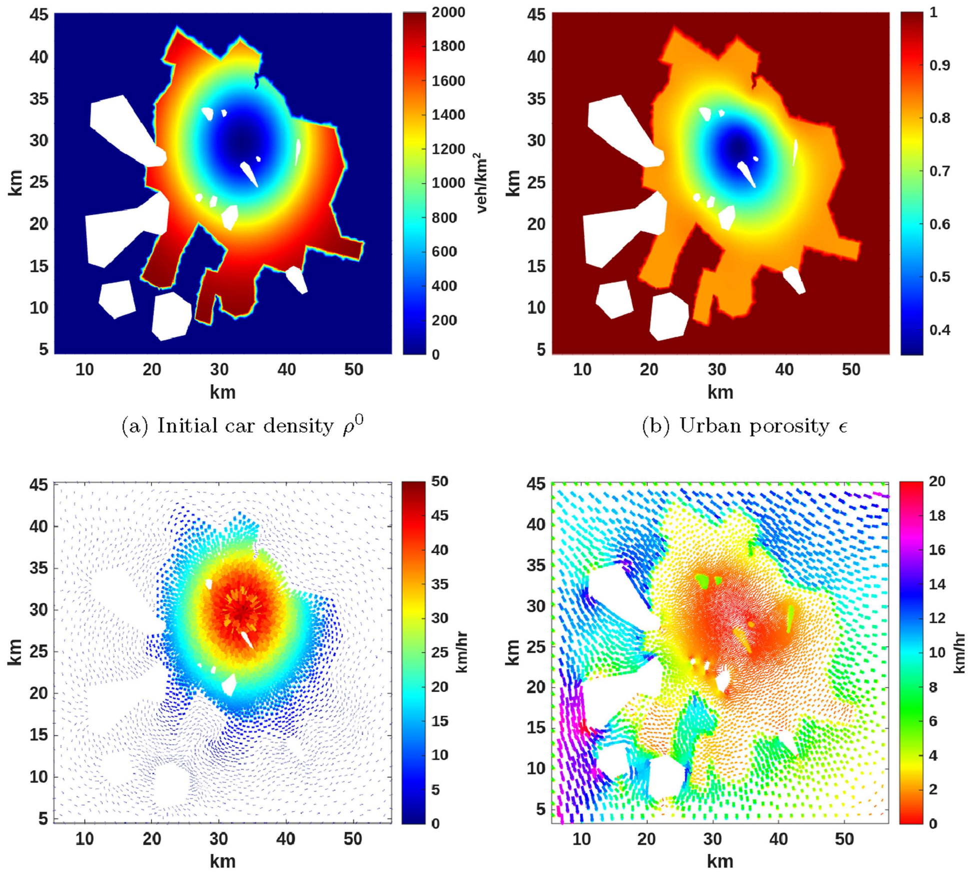
5.2 Initial values for the model variables

The numerical solution of above models requires well-defined initial conditions for the state variables and input data; these distributions are graphically represented across the domain Ω in Figure 3. Specifically, the initial traffic density $\rho^0(\mathbf{x})$, shown in Figure 3a, follows a concentric distribution where the maximum density



(a) Satellite image of the GMA (b) Triangular mesh and speed-limit zone

Figure 2. (a) Computational domain: the Guadalajara Metropolitan Area. Satellite image showing the urban domain, the rural surroundings, and natural obstacles (image taken from Google-Earth (2024)). (b) Triangular mesh generated with the software Gmsh, showing the subdomains, the boundary segments and also the speed-limit zone as a disk B of radius r for the single-objective problem (14).



(a) Initial car density ρ^0 (b) Urban porosity ϵ

(c) Initial desired velocity $\mathbf{u}_d$ (d) Wind field $\mathbf{v}$

Figure 3. Spatial distribution of initial states and given data: (a) Vehicle density distribution decreases from high values in the suburbs to nearly zero in the centre. (b) The urban landscape is assumed to be concentric; thus, the city exhibits high building density at the centre and more space for traffic flow in the suburbs. (c) By taking the initial density and fixing the city centre as an attraction point, the Eikonal problem generates a desired velocity pointing toward the centre; this velocity decreases as vehicle density grows. (d) A constant reference wind is assumed, blowing from the NW, with a noticeably lower velocity inside the city due to the urban porosity distribution.

is located in the suburban areas while the city centre remains nearly empty. This spatial configuration is characteristic of cities that have historically prioritized private vehicle flow over mass public transit. A clear example is Guadalajara (Mexico) which hosts over 5 million inhabitants and a motorization rate of nearly one vehicle for every two people, but has only three subway lines and three Bus Rapid Transit (BRT) lines.

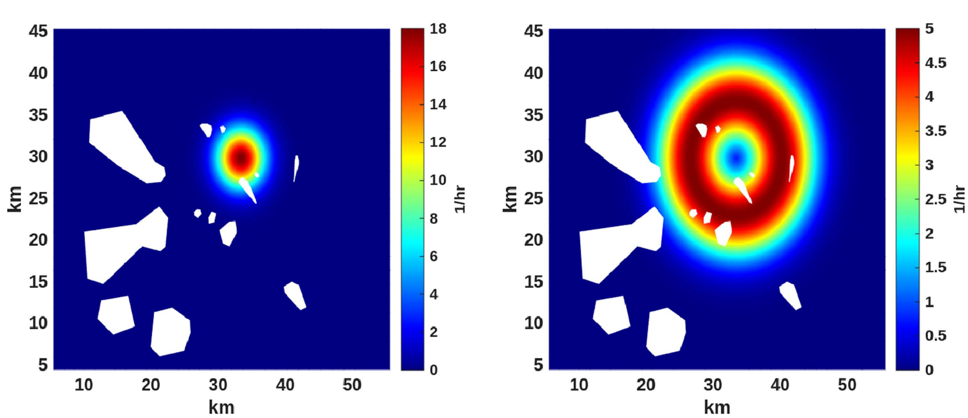
It is worth noting that due to the current absence of operational speed-limit and LEZ/ZEZs zones in the Guadalajara Metropolitan Area, this case study is configured as an ex-ante planning tool rather than an empirical post-hoc assessment. Accordingly, baseline traffic parameters as $\rho_{\max} = 2000$ veh/km^2, $\tau = 0.009$ h, $\nu = 1.25$ km^2/h, $\mu = 3.6 \times 10^{-8}$ km^2/h, $U_{\max} = 50$ km/h, and $\kappa_{\max} = 18$ h^{-1} are adopted from García Chan et al. (2025); Jiang et al. (2018), while the initial conditions are synthetically generated to demonstrate the feasibility of the optimization framework. However the model supports real-data parameters, even if they are not constants.

Complementing this, the urban landscape is characterized through the concept of urban porosity ϵ, following the concentric city model described in García-Chan et al. (2026). We consider a dense city with a concentric layout—a spatial structure that reflects typical monocentric urban sprawl and peripheral growth trends in emerging economies—where initial traffic and housing densities naturally decrease away from the city core. This peripheral expansion in Mexican cities has been heavily driven by housing finance policies that displaced low-income residential developments to outer urban rings while employment remained centralized (Guerra 2015; Monkkonen and Comandon, 2018). Accordingly, the urban medium is characterized by a baseline porosity value of $\epsilon = 0.38$ at the downtown core (Hu et al. 2012), which increases monotonically through the suburban rings until reaching a free-flow value of $\epsilon = 1.0$ in the rural surroundings (see Figure 3b). The interaction between these variables defines the desired velocity field, which is directed toward the city centre; it reflects near-zero speeds in congested zones and a free-flow maximum of 50 km/h, as illustrated in Figure 3c. Additionally, a reference wind field is generated by running the airflow model until a stationary state is reached. Rather than an arbitrary uniform vector, this stationary field operates under the Darcy–Brinkman–Forchheimer aerodynamic equations, directly capturing the spatial drag and physical resistance induced by the city's building blocks (see Figure 3d). Finally, all experiments are simulated assuming zero incoming traffic demand ($q = 0$) over a 2-h period to represent a morning rush hour scenario. This duration ensures that most vehicles shown initially present within the system either reach or approach their final parking destinations, enabling a comprehensive evaluation of traffic congestion and pollutant accumulation during peak hours.

The parking rate distribution may be fixed or variable depending on the specific optimal control problem. For the single-control problem (14), $\kappa(\mathbf{x})$ is fixed with a concentric distribution (see Figure 4a). In contrast, for the optimal control problem (20), the parking rate distribution depends on the radius r_h through the relation $\Delta_p = r_p - r_h$, where r_p is fixed. Consequently, the maximum parking rate values are located within a circular annulus of thickness Δ_p, which varies according to r_h (see Figure 4b).

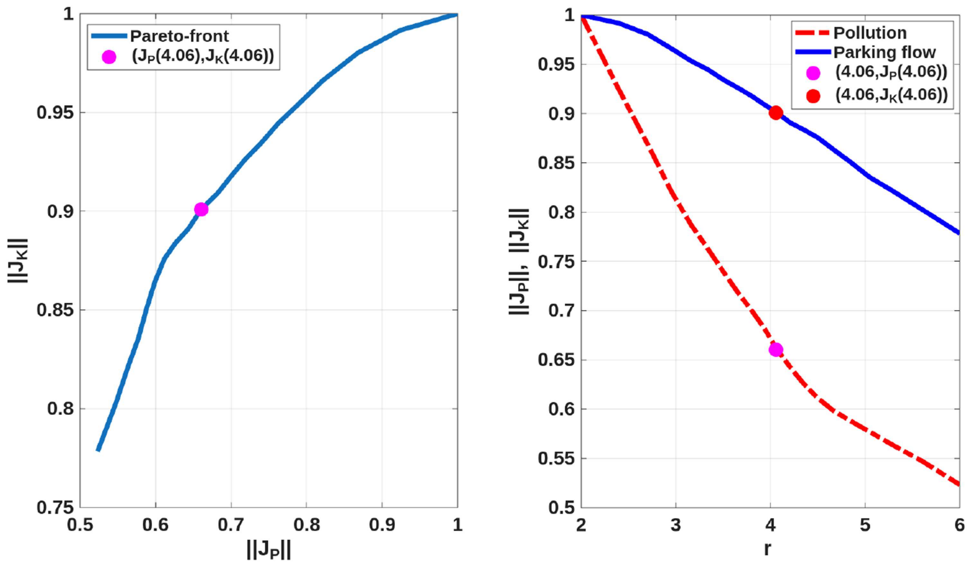
5.3 The optimal control problem 1

To solve the optimal control problem (14), we set the constraints $r_{lw} = 2$ and $r_{up} = 6$, and perform an exhaustive search to characterize the Pareto front. Taking advantage of the single-control nature of the radius r, a discrete vector $\mathbf{r} = [r_1, r_2, \ldots, r_m]$, with $m = 20$, was defined to evaluate the objective functionals across the entire admissible interval [2, 6]. This approach allowed for the identification of the maximum values $J_{P,\max}$ and $J_{K,\max}$, which were used to normalize the environmental and socioeconomic objectives, respectively. These normalized costs, $\|J_P\|_{\max}$ and $\|J_K\|_{\max}$, enable a direct comparison of the competing objectives as illustrated by the trade-off curve (Pareto front) in Figure 5a. This visualization reveals the set of non-dominated solutions, where any improvement in air quality J_P necessarily results in a degradation of the parking flow rate J_K. For instance, the front shows that prioritizing a parking rate greater than 0.95 restricts the reduction of pollution to levels no lower than 0.8; conversely, accepting a reduction in parking efficiency to 0.8 enables a significant environmental gain, lowering pollution to 0.6. Furthermore, Figure 5b displays the sensitivity of these costs to the control variable r. While the parking rate remains stable to increases in the radius, maintaining a value of 0.8 even at $r = 6$, the air pollution levels are sharply affected, decreasing by nearly half (0.55) at the same radius. Consequently, a compromise solution can be straightforwardly selected from this front based on the specific priorities and weights assigned by the decision-maker. As an illustrative example, in Figure 5 the results for the compromise solution corresponding to



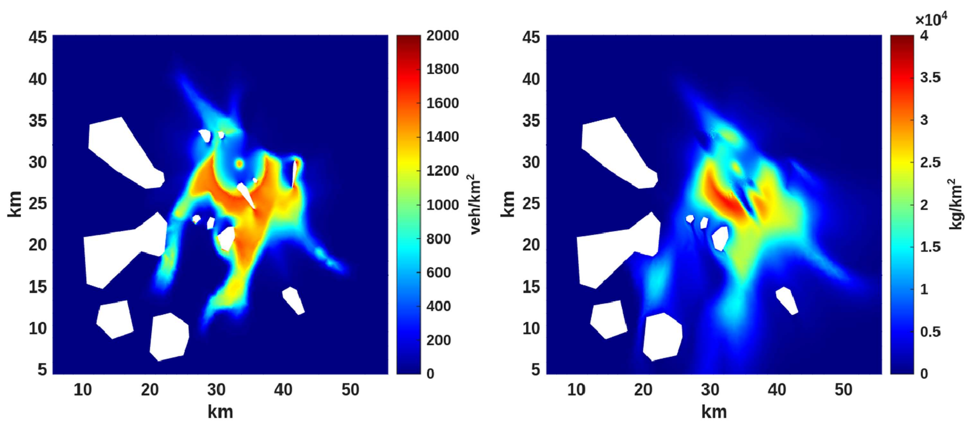
(a) Parking rate κ for problem 1 (b) Parking rate κ for problem 2

Figure 4. Parking rate distribution for the optimal control problems.



(a) Trade-off curve (b) Functional costs vs. radius

Figure 5. Characterization of the Pareto front and sensitivity analysis for the single-control problem.



(a) Car density (b) CO_2 concentration

Figure 6. Time-averaged distribution of state variables: (a) car density and (b) air pollution for the compromise solution $r = 4.06$ km, corresponding to the bullets shown on the trade-off curve (Figure 5).

$r = 4.06$ are highlighted with bullets, and its consequences on average cars density and air pollution are shown in Figure 6.

These graphical insights are quantified by the numerical data of Table 1, which presents the typical conflict found in environmental problems: maximizing commercial accessibility (parking flow) requires a smaller sensitive area, directly compromising air quality. In contrast, expanding the regulated radius moves the system towards the environmental extreme of the Pareto front.

5.4 The optimal control problem 2

Optimization problem (20) was solved using a hybrid derivative-free approach implemented in MATLAB 2025b via the `GA` and `Pattern Search` functions. To eliminate statistical dependency on random seeding and ensure an exhaustive search, the Genetic Algorithm was initialized using a population of 40 individuals. This initial population includes the operational lower and upper boundaries of the radii, while the

Table 1. Selected non-dominated solutions from the Pareto front for the single-control problem. The table presents both normalized and absolute values for the objective functionals J_P and J_K at various radii r, illustrating the quantitative trade-off between air quality and parking flow.

r	$\|J_P\|_{\max}$	$\|J_K\|_{\max}$	$J_P\,(\times 10^4)$	$J_K\,(\times 10^3)$
2.00	1.0000	1.0000	2.22	4.49
2.95	0.8208	0.9659	1.82	4.34
4.06	0.6605	0.9010	1.47	4.05
5.05	0.5750	0.8354	1.28	3.75
6.00	0.5233	0.7784	1.16	3.50

remaining 38 individuals are uniformly distributed across the admissible space but holding the fixed thickness Δr. The Genetic Algorithm was executed with a function tolerance of 10^{-2}, and to ensure a fair competition between the objectives, the cost functionals were normalized using maximum values estimated from the initial population. Subsequently, the best candidate generated by the GA was passed to the Pattern Search algorithm for local refinement, subject to a precision tolerance of 10^{-5}.

Regarding the physical configuration of the test cases, the cost parameters E_p, E_h, and E_e were set to unity due to the lack of empirical data, making implementation costs directly proportional to the area of each zone. Furthermore, the vehicle fleet is assumed to follow a distribution of $p_c = 0.7$, $p_h = 0.2$, and $p_e = 0.1$, representing an urban environment where internal combustion vehicles significantly outnumber their hybrid and electric counterparts. These parameters, along with the geometric constraints ($r_{h,lw} = 2.5$, $r_{h,up} = 6$, $r_{e,lw} = 0.5$, $r_{e,up} = 3$, and $\Delta r = 2$), define the search space for the four illustrative scenarios formulated by varying the weight vector $\boldsymbol{\lambda}_W = (\lambda_P, \lambda_K, \lambda_p, \lambda_h, \lambda_e)$ representing specific, predictable points on the Pareto front:

1. Unlimited resources *($\lambda_P = 1/2$, $\lambda_K = 1/2$, $\lambda_p = \lambda_h = \lambda_e = 0$)*: in this scenario, costs associated with urban modifications –particularly the establishment of new parking infrastructure outside the LEZ– are neglected. It is expected that, without this socioeconomic penalty, both radii expand to their maximum constraints. The outer radius r_h increases to displace combustion vehicles as far from the centre as possible, since the financial burden of accommodating them is unpenalized. Simultaneously, the inner radius r_e expands to enlarge the ZEZ. This keeps hybrid vehicles away from the core while strictly satisfying the LEZ annulus thickness constraint.
2. Limited resources ($\lambda_P = 1/3$, $\lambda_K = 1/3$, $\lambda_p = \lambda_h = \lambda_e = 1/9$): in this scenario, all implementation costs are equally penalized. We expect r_e and r_h to shift toward their minimum constraints to minimize the socioeconomic burden, thus reducing both the LEZ and ZEZ areas. This leads to higher pollution levels, as combustion cars remain closer to the centre. Furthermore, a minimized r_h reduces the area of the LEZ and, consequently, the space for parking infrastructure in the URZ, inherently limiting the maximum parking flow rate.
3. Parking-only *($\lambda_P = 1/3$, $\lambda_K = 1/3$, $\lambda_p = 1/3$, $\lambda_h = \lambda_e = 0$)*: this case evaluates the equilibrium between air quality and the costs proportional to the outermost parking area A_p associated with the LEZ. We anticipate a moderate increase in the LEZ outer radius r_h. However, the ZEZ inner radius r_e is also forced to increase in order to satisfy the geometric constraint $r_h - r_e \geq 2$ km, even if expanding the ZEZ area A_e is not economically penalized.
4. Electric-only *($\lambda_P = 1/3$, $\lambda_K = 1/3$, $\lambda_p = \lambda_h = 0$, $\lambda_e = 1/3$)*: this hypothetical case is designed specifically to force the optimization model to its extremes. By heavily penalizing the costs in the ZEZ area A_e, and neglecting them in the LEZ parking area A_p, the model drives the LEZ outer radius r_h towards its maximum bound in order to improve air quality. Simultaneously, the ZEZ inner radius r_e contracts to its minimum allowable limit to avoid the severe costs associated with the only-electric zone.

Remark 1. The normalization of the unit implementation costs ($E_p = E_h = E_e = 1$) does not structurally distort the identification of the Pareto-optimal points. In Cases (i), (iii), and (iv), the weights assigned to these cost objectives are either zero or isolated, meaning they do not actively compete. Conversely, in Case (ii), where a uniform weight distribution is considered, the optimization effectively triggers a direct trade-off between environmental benefits against the total spatial covered by the restricted zones.

The application of aforementioned hybrid strategy to solve these four test cases yields a highly efficient optimization process. The convergence profile in Figure 7 confirms that the GA effectively identifies a global candidate region where population fitness stabilizes, allowing the PS stage to finalize the search with high precision at a much lower computational cost than GA.

Having established the numerical efficiency of the solver, we now examine the physical implications of the optimized scenarios. Across all test cases, the vehicle density maps demonstrate a clear spatial segregation of the fleet according to the imposed restrictions. Internal combustion vehicles, representing the majority of the traffic, are effectively restricted to the region outside the LEZ, leading to visible accumulation along its boundaries. Conversely, hybrid vehicles are allowed to penetrate the LEZ but remain excluded from the ZEZ. As a result, the ZEZ exhibits a near-zero vehicle density in all simulations. This traffic distribution directly dictates the CO_2 concentration patterns. The ZEZ remains nearly free of pollution, with only trace concentrations present due to atmospheric transport by wind fields from neighbouring areas. The LEZ also exhibits significantly low pollution levels, as it only accounts for the reduced emissions of hybrid vehicles –modelled here with a 50% reduction relative to combustion engines– and the influx of transported pollutants. Consequently, the highest CO_2 concentrations are strictly confined to the URZ, where the bulk of the combustion-based fleet is concentrated.

Building upon this general traffic behaviour, we now analyze the specific Pareto-optimal configurations for each scenario: The optimal solutions summarized in Table 2 align strictly with our predictions. In the 'Unlimited resources' scenario, the ZEZ and the LEZ expand upto their maximum constraints. This effectively shields the city core from pollution and avoids congestion by pushing the URZ outward, where wide parking infrastructure can be accommodated (see Figure 8a and b). In contrast, in the 'Limited resources' scenario, the quadratic growth of area-based costs outweighs the marginal environmental gains, forcing the ZEZ and LEZ radii (r_e, r_h) towards their lower bounds. As shown in Figure 8c and d, this fact limits the outward displacement of the URZ, leading to insufficient parking area for a higher car density and, subsequently, generate higher congestion –and consequently more travel time– and higher pollution levels near the city core area.

The 'Parking-only' scenario reflects a balanced urban intervention, featuring a moderate expansion of both boundaries relative to their constraints. Expanding the LEZ outer radius r_h simultaneously forces the ZEZ inner radius r_e to expand in order to maintain the 2 km minimum thickness. This incurs no additional

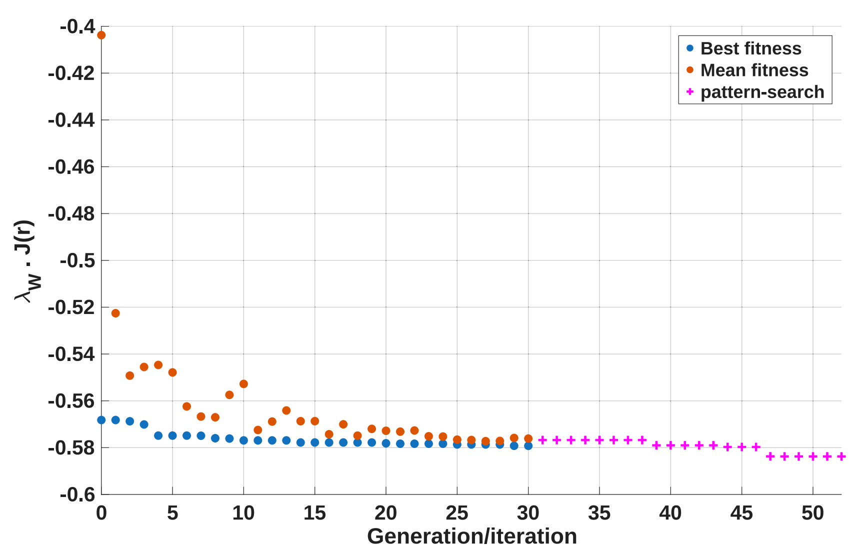
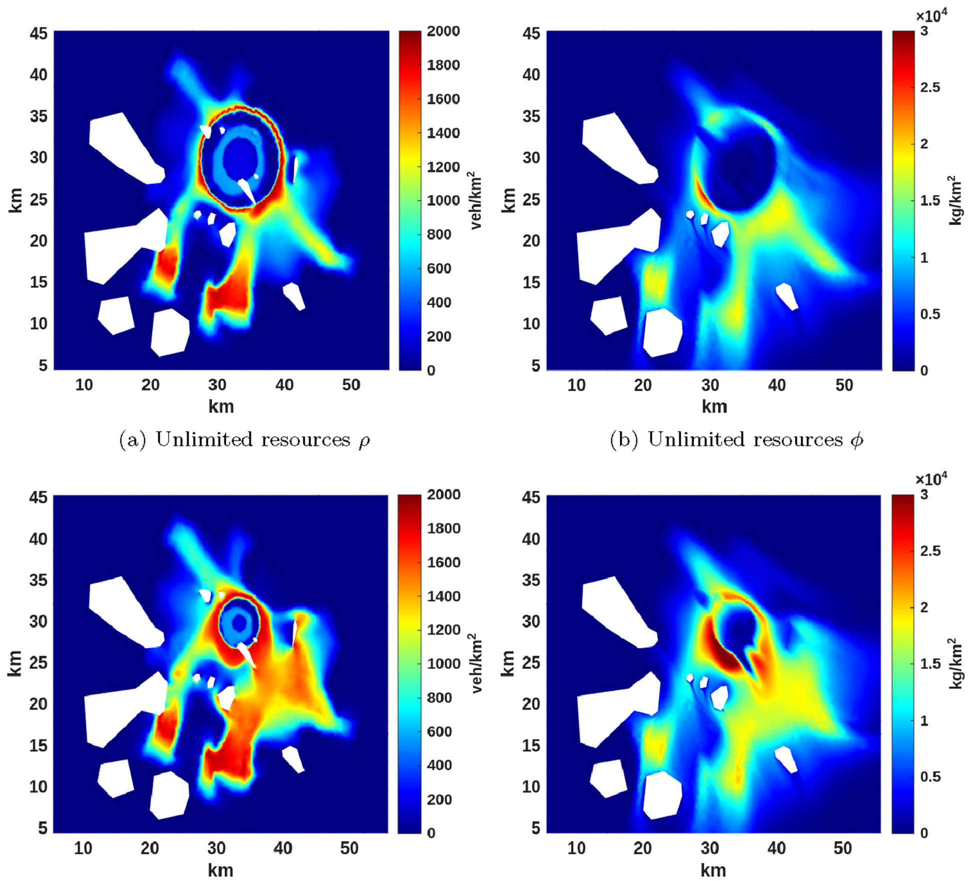


Figure 7. Convergence profile of the hybrid solver. The plot shows the evolution of the objective functional, distinguishing the GA phase from the PS refinement. The GA stage concludes when the best and the mean fitness values (blue and red dots, respectively) converge, providing an optimal seed for the PS (magenta dots) to achieve a high quality final solution.

Table 2. Comparison of Pareto-optimal solutions across different scenarios for problem 2. The objective functionals J_P and J_K represent the normalized environmental and mobility costs, respectively. The terms A_p, A_h, and A_e denote the area covered by the URZ, LEZ and ZEZ (km^2). Each scenario corresponds to a specific configuration of the weight vector $\boldsymbol{\lambda}_W$, illustrating the trade-offs between urban infrastructure investment and environmental quality.

Case scenario	r_e	r_h	$J_P\,(\times 10^3)$	$J_K\,(\times 10^3)$	A_p	A_h	A_e
(i) Unlimited res.	2.5000	5.8552	3.47	1.53	39.93	88.06	19.63
(ii) Limited res.	0.5000	3.0000	7.09	1.42	21.99	27.49	0.79
(iii) Parking-only	2.0712	4.1254	5.12	1.46	29.06	39.99	13.48
(iv) Electric-only	0.5037	5.8547	3.67	1.54	39.98	106.89	0.79



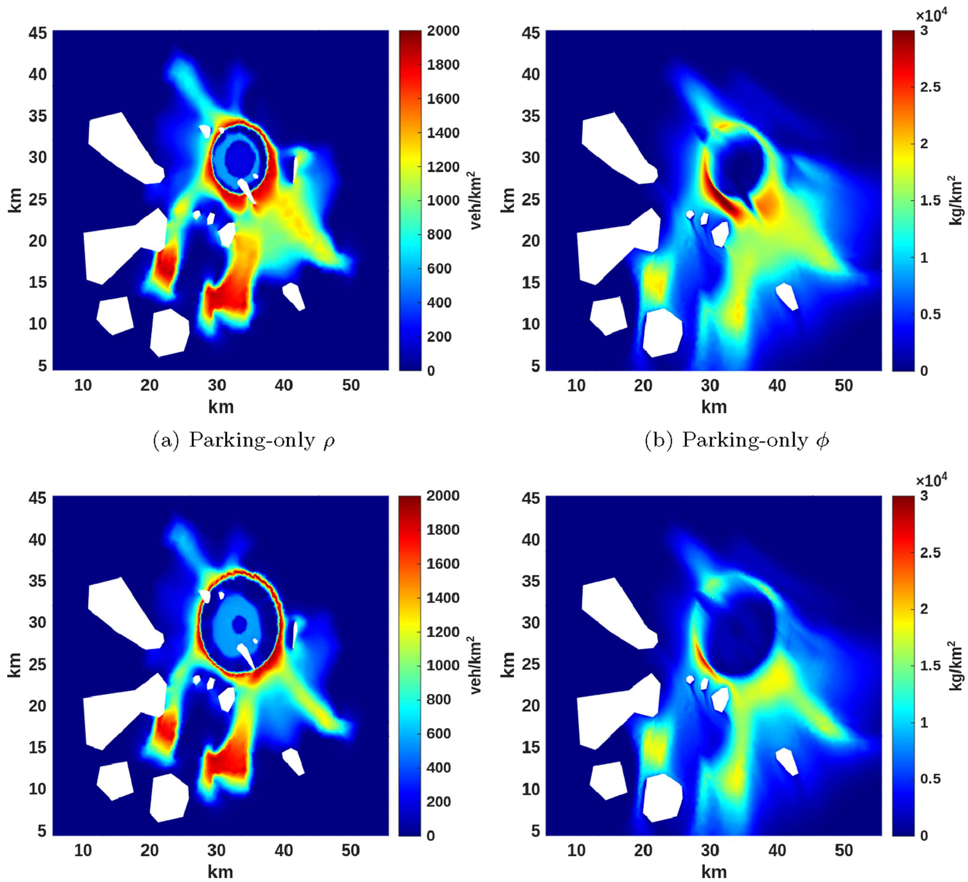
(a) Unlimited resources ρ (b) Unlimited resources ϕ

(c) Limited resources ρ (d) Limited resources ϕ

Figure 8. Time-averaged spatial distribution of state variables: The left column displays car density ρ, while the right column shows CO_2 concentration ϕ for the first two optimal scenarios listed in Table 2. From top to bottom, the rows correspond to cases: (i) unlimited resources, (ii) limited resources.

penalty since the ZEZ area costs are neglected ($\lambda_e = 0$). As shown in Figure 9a and b, this configuration yields a cleaner urban core while avoiding severe congestion in the URZ. Meanwhile, the ‘Electric-only’ scenario serves as a theoretical stress test for the solver's consistency. Here, the ZEZ radius r_e contracts to its minimum limit to avoid heavily penalized infrastructure costs. Concurrently, the LEZ radius r_h expands, acting as a buffer that pushes combustion vehicles outward into the URZ. Consequently, while the worst emissions are kept far from the core, the confined ZEZ allows hybrid vehicles closer to the centre, leading to a slight localized increase in pollution (see Figure 9c and d).

To further demonstrate the sensitivity of the LEZ/ZEZ zones to fleet composition, we extend the analysis of the 'Parking-only' scenario defined by the weights $\lambda_P = 1/3$, $\lambda_K = 1/3$, $\lambda_p = 1/3$, $\lambda_h = 0$, $\lambda_e = 0$, considering the cleaner vehicle penetration trends in Mexico (Islas-Samperio, Manzini, and Grande-Acosta 2020). Specifically, we execute the algorithm assuming a plausible mid-term composition fleet of $p_c = 0.55$, $p_h = 0.3$, $p_e = 0.15$. Our



(a) Parking-only ρ (b) Parking-only ϕ

(c) Electric-only ρ (d) Electric-only ϕ

Figure 9. Time-averaged spatial distribution of state variables: the left column displays car density ρ, while the right column shows CO_2 concentration ϕ for the last two optimal scenarios listed in Table 2. From top to bottom, the rows correspond to cases: (iii) parking-only, (iv) electric-only.

results show that the Pareto-optimal radii are contracted to $r_e = 1.7281$ and $r_h = 3.7847$ (down from $r_e = 2.0712$ and $r_h = 4.1254$). This geometric contraction is due to the fact that less combustion vehicles produce less tailpipe emissions mitigating the average CO_2 concentration ($J_P = 5.06 \times 10^3$) on the restriction zone. Consequently, the demand of parking spaces is relaxed ($J_K = 1.34 \times 10^3$) and then shrinks its cover area ($A_p = 26.9212$). Furthermore, since $\lambda_h = \lambda_e = 0$ implies zero penalization for allocating hybrid and electric vehicles, the algorithm directly compresses the remaining internal zones to $A_h = 35.6192$ and $A_e = 9.3820$.

6 Conclusions

This work presents a novel methodology for the design and delimitation of restricted urban zones through a multi-objective optimization framework. The approach identifies Pareto-optimal solutions that balance the minimization of average pollutant concentrations with the maximization of parking flow, utilized here as a proxy for socioeconomic activity and traffic congestion. In order to maintain generality and computational efficiency, the radii of concentric zones serve as the primary control variables.

Two distinct urban interventions were evaluated: a speed-limit zone (Zone 30) and a graduated prohibition strategy (LEZ and ZEZ). The latter implements a two-stage filter that restricts internal combustion vehicles in the outer annulus and excludes hybrid vehicles from the inner disk, effectively reserving the city core for only-electric mobility. The numerical experiments focused on CO_2 emissions, but can be directly extended to other health-critical pollutants like NO_2, $PM_{2.5}$ or VOCs, because these pollutants share a common tailpipe origin and, therefore, any policy implemented to promote a reduction in fuel consumption produces a decrease in air toxicity.

The results from the first optimization problem, characterized by the Pareto front, reveal that minimizing pollution requires expanding the radius of the limited area, which in turn reduces the inflow rate of vehicles accessing the parking infrastructure. By generating these trade-off curves, this study provides the set of non-dominated solutions, offering decision-makers a tool to select an operating point based on specific policy priorities.

Regarding combined LEZ/ZEZ zones, the consistency of the framework was evaluated across four benchmark scenarios using the weighted sum method. The model's ability to converge towards predictable optimal outcomes in these test cases confirms its reliability in identifying specific Pareto-optimal points. This flexibility allows urban planners to prioritize different aspects of the urban transition—such as infrastructure costs versus air quality—by simply adjusting the weight vector λ_W. Furthermore, our numerical simulations suggest that LEZ/ZEZ boundaries are not rigid but adapt to fleet composition, where the mid-term growth of hybrid and electric vehicles naturally leads to a contraction of the restricted areas. This underscores a public policy insight: rethinking the trade-off between allocating fiscal incentives for private fleet renewal versus financing costly urban infrastructure interventions in parking spaces and public transportation.

Finally, the nature of this macroscopic framework is not capable of incorporating microscopic elements widely used in traffic flow management, such as traffic light timing and lane management. Furthermore, the lack of calibration for continuous parameter—such as urban porosity—against local datasets, combined with the current absence of operational restrictive zones in the city of Guadalajara, renders this work a theoretical baseline. Nevertheless, once these limitations are addressed in future stages, this framework has the potential to become a versatile decision-making tool for evaluating urban interventions before full-scale implementation.

Acknowledgements

We thank the anonymous reviewers for their comments that helped to improve this paper.

Author contributions

CRediT: **Lino J. Alvarez-Vázquez:** Conceptualization, Investigation, Methodology, Visualization, Writing – review & editing; **Aurea Martínez:** Conceptualization, Investigation, Methodology, Visualization, Writing – original draft; **Miguel E. Vázquez-Méndez:** Conceptualization, Investigation, Methodology, Visualization, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Ministerio de Ciencia y Tecnología [Grant No. TED2021-129324B-I00]; Secretaría de Ciencia, Humanidades, Tecnología e Innovación [Grant No. CBF-2025-I-380]; PRODEP [Grant number 103/16/8066]; Sistema Nacional de Investigadores [Grant No. 52768].

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